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Data Analysis

Final assignment

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Introduction

We analyzed **Rossmann Store** Sales found on Kaggle: link

The dataset has:

1. *1115 stores*
2. *750-980 rows* for each store.
3. Each row consists of *Sales, Promotions, Customers*, etc.

Store	DayOfWeek	Date	Sales	Customers	Open	Promo	StateHoliday	SchoolHoliday
1	5	31/07/2015	5263	555	1	1	0	1
2	5	31/07/2015	6064	625	1	1	0	1
3	5	31/07/2015	8314	821	1	1	0	1

Dataset shape: 1,017,209 x 9

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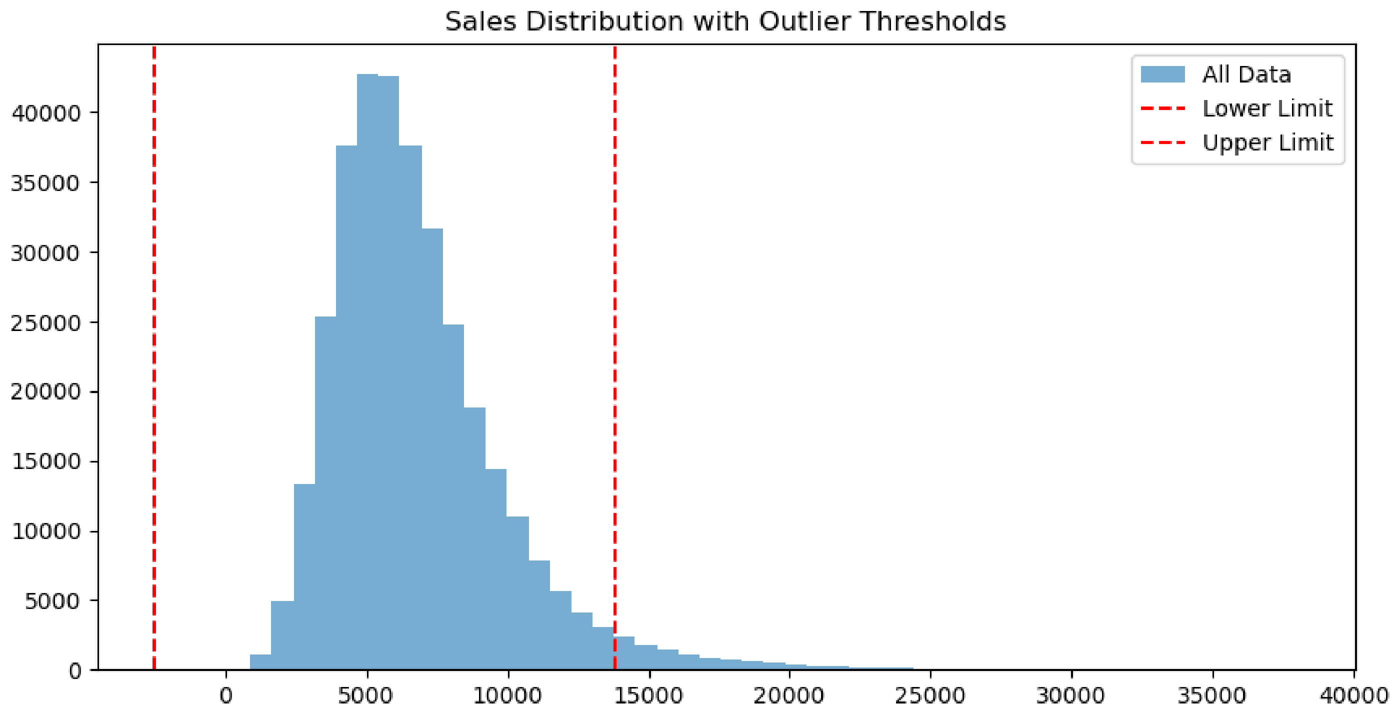
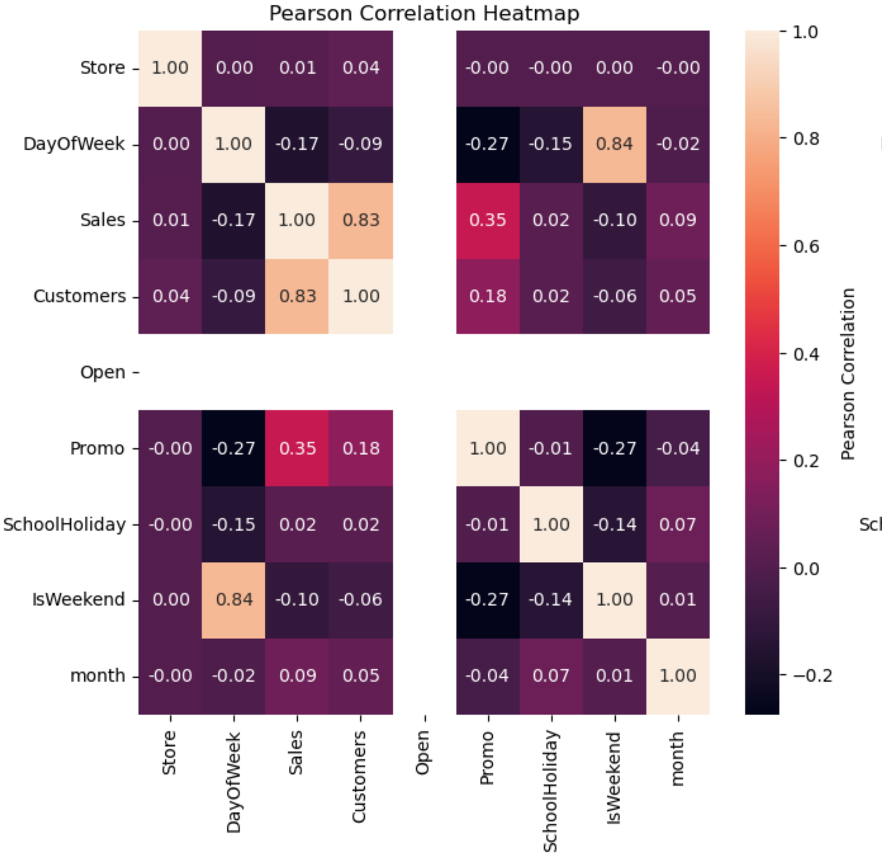
Data preprocessing

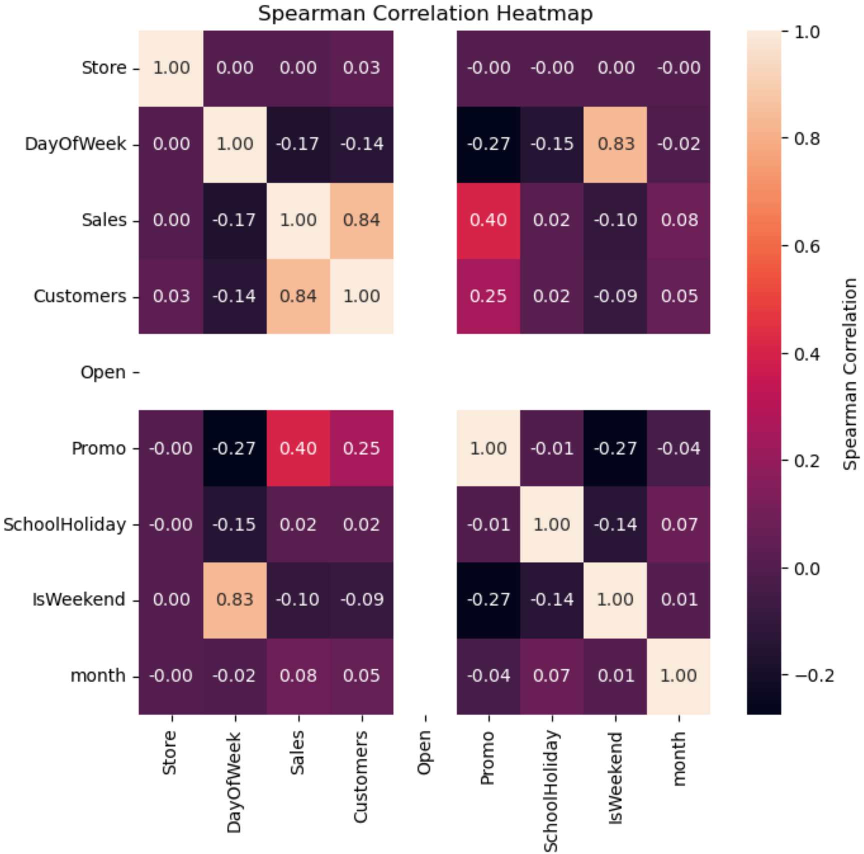
1. Extracted: 406,974 rows
2. Null values: 0 cells
3. Duplicates: 0 rows
4. No sale days: 69,050 (removed)
5. Outliers: 11,167 (not removed)
6. Features engineered: “IsWeekend”

Dataset shape: 337,924 x 10

Outliers


Sales Distribution with Outlier Thresholds
All Data
Lower Limit
Upper Limit
0
5000
10000
15000
20000
25000
30000
35000
40000





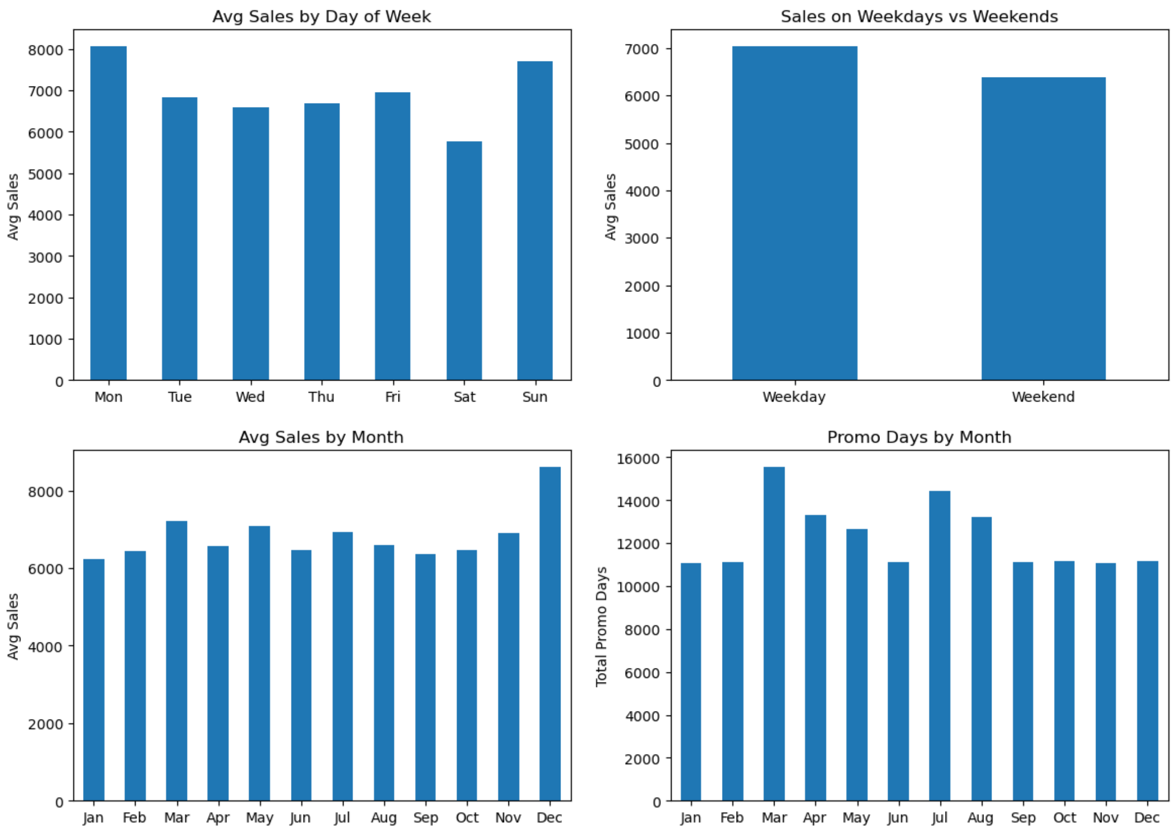
Pearson Correlation Heatmap

	Store	DayOfWeek	Sales	Customers	Open	Promo	SchoolHoliday	IsWeekend	month
Store	1.00	0.00	0.01	0.04		-0.00	-0.00	0.00	-0.00
DayOfWeek	0.00	1.00	-0.17	-0.09		-0.27	-0.15	0.84	-0.02
Sales	0.01	-0.17	1.00	0.83		0.35	0.02	-0.10	0.09
Customers	0.04	-0.09	0.83	1.00		0.18	0.02	-0.06	0.05
Open									
Promo	-0.00	-0.27	0.35	0.18		1.00	-0.01	-0.27	-0.04
SchoolHoliday	-0.00	-0.15	0.02	0.02		-0.01	1.00	-0.14	0.07
IsWeekend	0.00	0.84	-0.10	-0.06		-0.27	-0.14	1.00	0.01
month	-0.00	-0.02	0.09	0.05		-0.04	0.07	0.01	1.00

Spearman Correlation Heatmap

	Store	DayOfWeek	Sales	Customers	Open	Promo	SchoolHoliday	IsWeekend	month
Store	1.00	0.00	0.00	0.03		-0.00	-0.00	0.00	-0.00
DayOfWeek	0.00	1.00	-0.17	-0.14		-0.27	-0.15	0.83	-0.02
Sales	0.00	-0.17	1.00	0.84		0.40	0.02	-0.10	0.08
Customers	0.03	-0.14	0.84	1.00		0.25	0.02	-0.09	0.05
Open									
Promo	-0.00	-0.27	0.40	0.25		1.00	-0.01	-0.27	-0.04
SchoolHoliday	-0.00	-0.15	0.02	0.02		-0.01	1.00	-0.14	0.07
IsWeekend	0.00	0.83	-0.10	-0.09		-0.27	-0.14	1.00	0.01
month	-0.00	-0.02	0.08	0.05		-0.04	0.07	0.01	1.00

Daily/Periodic Pattern Analysis of all stores



Probability analysis

```
P(HighSales | Promo=1): 18.03 %
P(HighSales | Promo=0): 3.87 %

P(LowSales | Promo=1): 0.82 %
P(LowSales | Promo=0): 17.42 %
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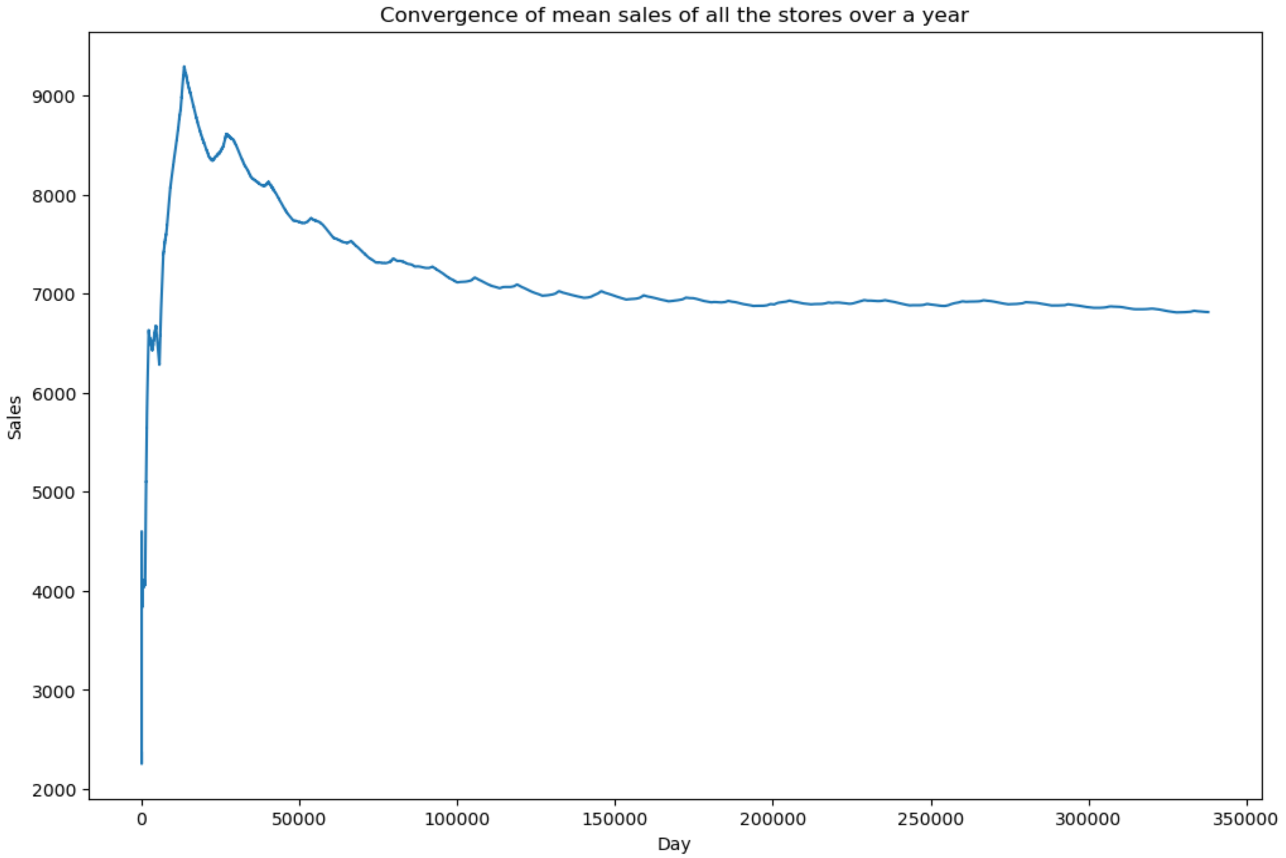
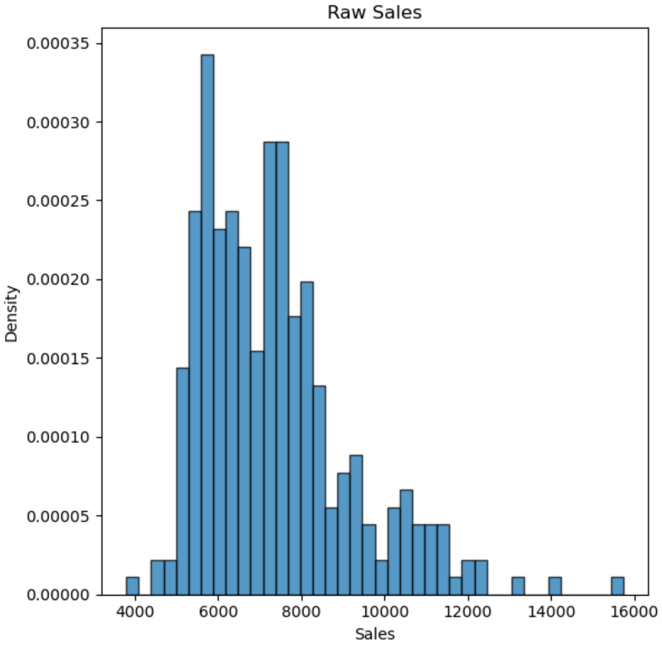
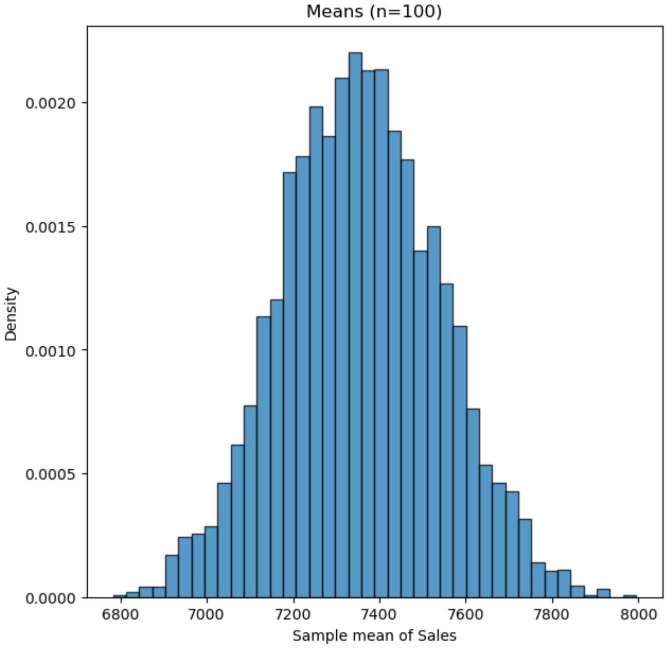
IsWeekend	0	1
Promo		
0	54.97	45.03
1	80.45	19.55

Law of Large Numbers



Central Limit Theorem





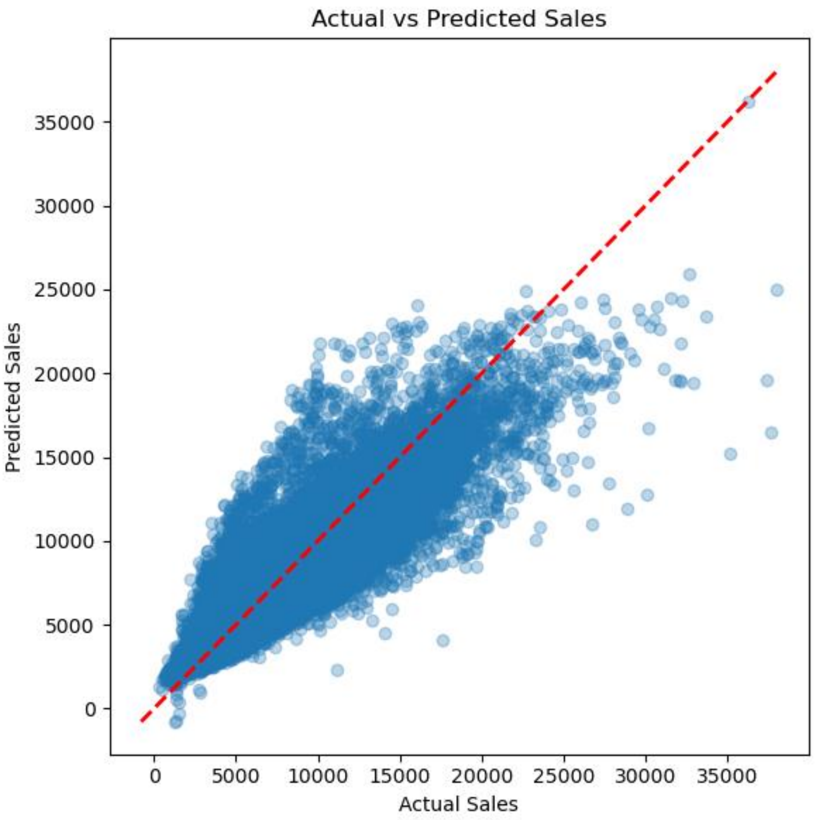
Actual vs. Predicted sales

Mean sales: 6814.95
Root mean square error (polynomial-2 linear regression): 1497.67 (21.97%)

Inaccuracy is likely due to:
1. Calendar patterns
2. Sufficiency problems
3. Varying promotions

R^2: 0.77
MAE: 1076.48

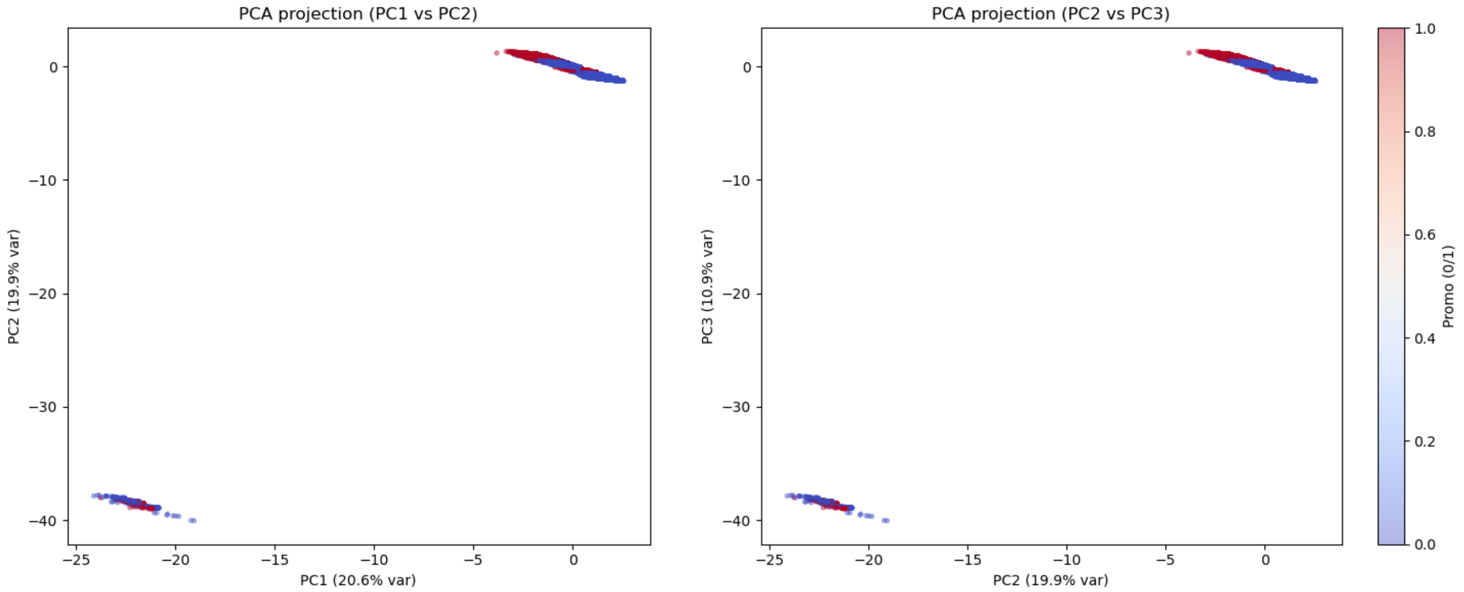
Reference notebooks were searched and algorithms such as Random Forest, XGBoost show accuracy reaching ~90% instead of ~83% here.



PCA

Explained variance ratio:

PC1: 20.618%
PC2: 19.853%
PC3: 10.908%



PCA

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Cumulative variance:

k=2: 40.47%
k=3: 51.38%

k for 80%: 6
k for 90%: 7
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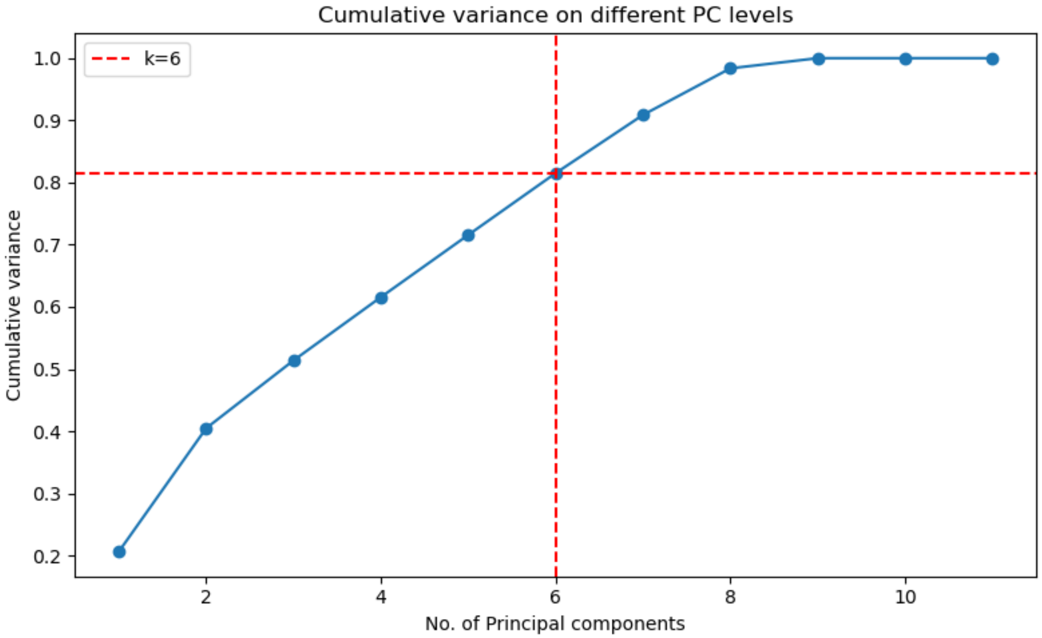

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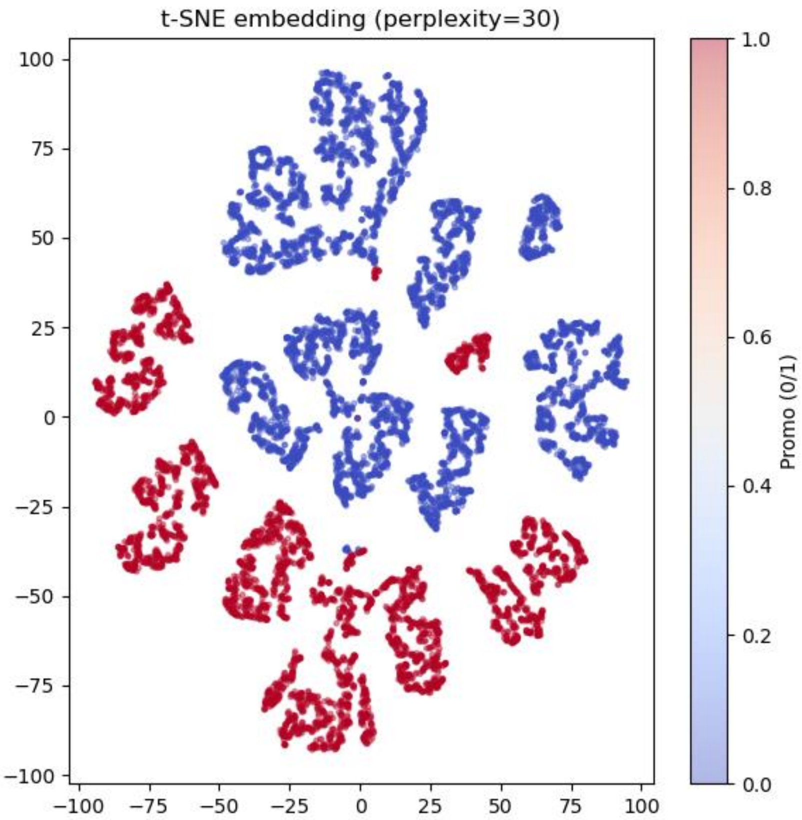
t-SNE

- Since we see that Promo regions are different from non-promo regions, there's a fundamental difference between the two.



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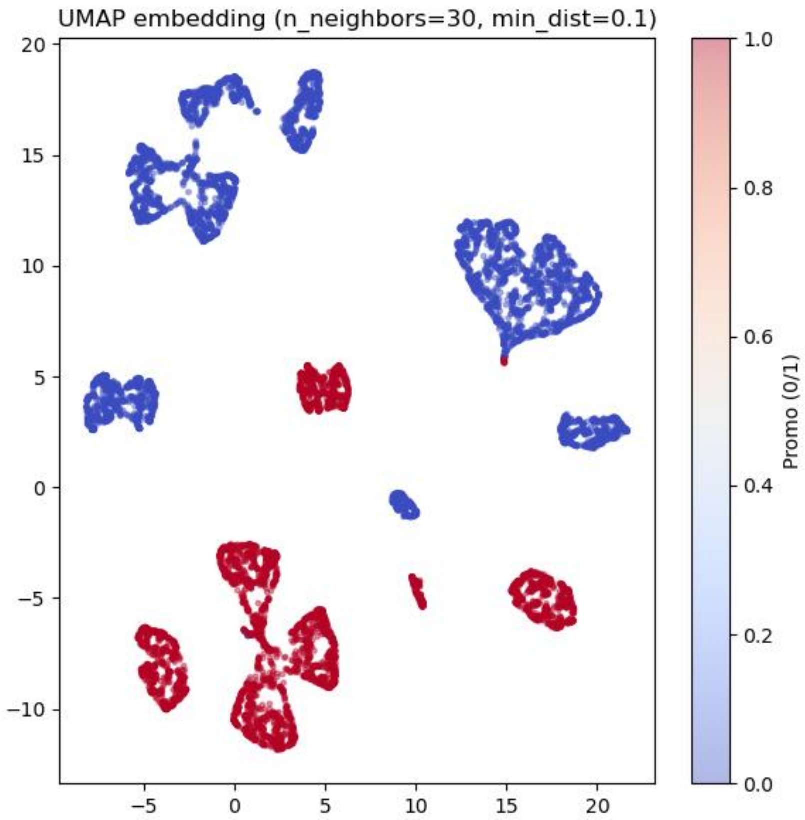
UMAP

- We observe the local neighborhoods being preserved.



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Chi-square test

Null Hypothesis: The events "IsWeekend" and "Promo" are independent.

Results for the test:
chi2=24192.940
dof=1
p-value=0

Hypothesis rejected, explanation:
Chances of Promo happening on a weekend: 20%.
Chances of Promo happening on a weekday: 45%.

IsWeekend	0	1
Promo		
0	104492	86444
1	118060	28928

QUESTIONS?